TAMIZHAZHAGAN S

ECE Engineer · Full-Stack Java Developer · IoT Systems Researcher

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QUICK LINKS

- Portfolio
- LinkedIn
- GitHub
- HackerRank
- LeetCode

TECHNICAL SKILLS

Languages
Java · C · Embedded C · Python · JavaScript · SQL

Web / Backend
HTML5 · CSS3 · JDBC · REST APIs · MySQL · OOP · MVC

Embedded / IoT
Arduino Nano/Uno · MPU6050 · NEO-6M GPS · SIM800L · AT Commands · Sensor Fusion

EDA / Tools
KiCad · Proteus · Multisim · Eagle · Arduino IDE · VS Code · IntelliJ · Git · Linux

Problem Solving
LeetCode 35+ (Hard) · HackerRank 12 Certs · DSA & Algorithms

EDUCATION

B.E. – ECE
AVS Engineering College
Anna University, Salem
2023 – 2026

Diploma – EEE
Thiagarajar Polytechnic College
Salem
2020 – 2023

SSLC
Govt. Higher Secondary School
Singipuram
2019 – 2020

HACKERRANK CERTS

- Software Engineer (Full-Stack)
- Software Engineer Intern
- Java (Basic)
- JavaScript (Basic & Intermediate)
- Python (Basic)
- CSS (Basic)
- SQL (Basic · Intermediate · Advanced)
- Problem Solving (Basic & Intermediate)

ACHIEVEMENTS

- Published – IJEETR Mar–Apr 2026
- 3rd Prize – Project Expo, Muthayammal College (Dec 2024)
- Presenter – VARNAM'24, KIT (Dec 2023)

PROFESSIONAL SUMMARY

B.E. ECE graduate (Anna University, 2023–2026) with **peer-reviewed research published in IJEETR 2026** and hands-on Java internship experience. Builds end-to-end systems — from **100 Hz embedded C on microcontrollers** to production Java backends and analog RF circuits. 12 verified HackerRank certifications and 35+ LeetCode problems (including Hard tier). Actively seeking roles in embedded systems, IoT engineering, or full-stack Java development.

INTERNSHIP EXPERIENCE

Full Stack Development Intern · *Industry Internship* **Sep – Oct 2025**

- Built responsive web applications integrating HTML/CSS/JS frontend with Java & MySQL backend using RESTful API principles.
- Optimised SQL queries and UI performance; delivered sprint-cycle features using Git-based collaborative workflows.

Java Development Intern · *Industry Internship* **2023**

- Gained real-world Java development experience on a production codebase with JDBC database integration and OOP design patterns.
- Implemented professional software development workflows: version control, code review, and structured project delivery.

RESEARCH & PUBLICATION

IoT-Based Smart Helmet with Accident Detection & Emergency Alert System
IJEETR, Vol. March–April 2026 · ***Peer-Reviewed & Published***

- Designed a life-critical, fully offline embedded system achieving sub-5s end-to-end emergency response for motorcycle accident victims.
- 100 Hz IMU sampling (MPU6050) → crash-threshold calibration → GPS fetch (NEO-6M) → GSM SMS alert (SIM800L) — zero internet dependency.
- ₹3,500 BOM; peer-reviewed and published; team of 4, 6-month development cycle — production-ready hardware prototype.

PROJECTS

BookMyShow Ticket Booking System
Java · JDBC · MySQL · OOP · MVC

- Built a full-stack Java desktop application with dual-role authentication (admin + user) and role-based access control.
- Implemented JDBC-backed transactional seat locking with SELECT FOR UPDATE to eliminate double-booking race conditions.
- Designed clean MVC architecture with zero code duplication and a robust SQL schema enforcing referential integrity.

FM Spy Bug — Wireless FM Transmitter
RF Electronics · Analog Circuit Design · FM Modulation

- Designed a discrete-component FM transmitter — no microcontroller — with optimised oscillator, antenna geometry, and impedance matching.
- Demonstrated mastery of analog RF fundamentals: oscillation, frequency modulation, and impedance matching on standard receivers.

Automatic Industrial Cleaning Bot
Robotics · Embedded Systems · Ultrasonic · Motor Control · Wireless RF

- Built an autonomous mobile robot with ultrasonic obstacle detection, differential drive control, and integrated cleaning mechanism.
- Implemented wireless remote override and repeatable cleaning path logic; awarded 3rd Prize at Project Expo, Muthayammal College (Dec 2024).

Smart Appointment Scheduler
HTML · CSS · JavaScript · MySQL · Full-Stack

- Full-stack web application with scheduling, CRUD operations, and reminder system using HTML, CSS, JavaScript, and MySQL backend.